

F: Heist

Source file name: `heist.c`, `heist.cpp`, `heist.java`, or `heist.py`

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Danny O'Sea is pulling the biggest, most awesome heist of her life. *"This is the last one"* she promised you and your crew, *"after this one I will retire"*.

You and your team of expert hackers / infiltrators, of course, know better, but what can you do? The pay is so good.

*"Here's the plan: the **Ultra Luxurious McGuffin** is currently displayed in the **Very Secure Gallery**, in the **Extremely Securest Room**, which, as expected, is protected by a set of lasers that will trigger an alarm if interrupted.*

*"Now, I have acquired, at a great cost, a list of the lasers' positions in the room. So, what I need you to do is figure out what is the **safest location** in the room. We're pulling an *Ethan Hunt*".¹ She winks at you.*

You look at the map of the **Extremely Securest Room**. It is a rectangle of $W \times L$, dangerously criss-crossed with a list of Z lasers. Each laser is defined by its emitter and its receiver, two points $E = (ex, ey)$ and $R = (rx, ry)$, both of which lie on the walls of the room.

As you may know, from any heist movie, or well, your actual experience, you need to be as far as possible from any laser.

"Don't forget the walls" - says Danny - *"stay away from the walls."* Ok, as far as possible from any laser and any wall.

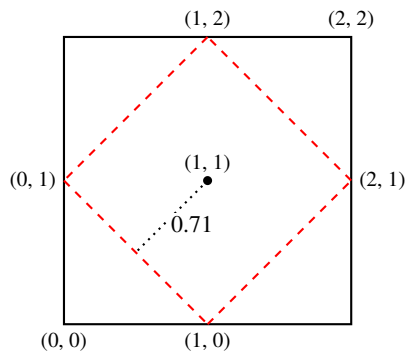
Your mission is to find the safest point to land in the Extremely Securest Room, that is, the point that is farthest from any laser path, and any wall.

"And the distance to the closest laser, or wall" - says Danny - *"we need to know the distance!"* Ok, the safest point and the distance to the closest laser or wall.

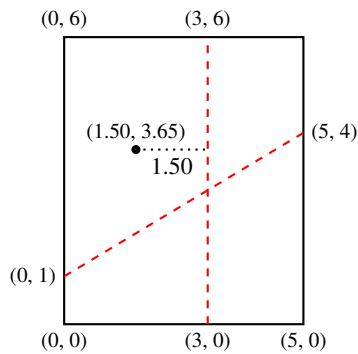
If more than one point is farthest from any laser path or wall, pick the one that lies in the largest area. If more than one such point lies in the largest area, or there are more than one largest areas, pick the one closest to the coordinate $(0, 0)$, why? Because that's the emergency exit.

Here are some examples:

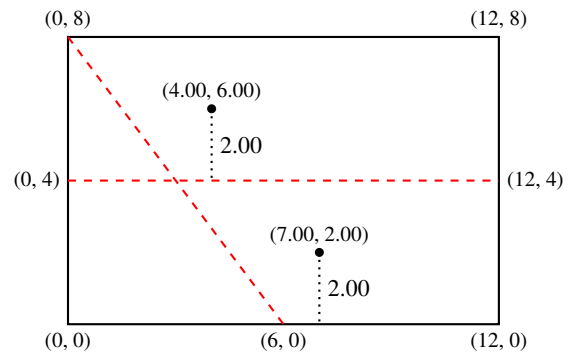
¹This is heist lingo for "hanging from the ceiling, snatching the actual prize, and pulling back"



Example 1



Example 2



Example 3

- On Example 1, it's evident the point in the center is farthest from all lasers and walls.
- On Example 2, there are multiple safest points in the larger region, the one depicted is the closest to (0, 0).
- On Example 3, there are multiple safest points in both the top and below areas, the one at the top is on the largest area, and closest to (0, 0).

Input

The input consists of several test cases. Each test case starts with three integers W, L, Z , ($1 \leq W, L \leq 1000$), ($1 \leq Z \leq 100$) separated by single spaces: W and L are the dimensions of the **Extremely Securest Room**. Z is the number of lasers installed.

Each of the following Z lines contains four integers ex, ey, rx, ry , ($0 \leq ex, rx \leq W$), ($0 \leq ey, ry \leq L$) separated by single spaces, which are the positions of the **laser emitter** and its corresponding **receiver**, which are each on a different wall of the room.

The last line of input will be `0 0 0` and should not be processed.

The input must be read from standard input.

Output

For each test case, print a single line with three decimal numbers sx, sy, d , the position of the **safest location** to land in the room, and the distance to the closest laser or wall, with two decimals, and an error of at most 10^{-2} .

The output must be written to standard output.

Sample Input	Sample Output
2 2 4	1.00 1.00 0.71
1 0 2 1	1.50 3.65 1.50
2 1 1 2	4.00 6.00 2.00
1 2 0 1	
0 1 1 0	
5 6 2	
3 0 3 6	
0 1 5 4	
12 8 2	
0 4 12 4	
0 8 6 0	
0 0 0	